

Oil Price Shocks and Democratization: Evidence from Regime-Dependent Effects

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Abstract

This paper revisits Brückner, Ciccone, and Tesei (2012), who find that positive oil price shocks generate improvements in democratic institutions. We ask whether this relationship is conditional on countries' initial political institutions and whether oil booms and busts have symmetric effects across regime types. Using their replication data, we first reproduce their baseline results and then allow the effect of oil shocks to vary nonlinearly with initial democracy. An endogenous threshold model identifies a regime boundary around a Polity2 score of -6 , although the associated confidence interval is wide. Above this threshold, oil price shocks are positively associated with subsequent changes in democracy, while below it the estimated effect is small and statistically insignificant. Local projections further reveal substantial asymmetry in the response to oil booms and busts: hard autocracies show little response to positive oil shocks but experience pronounced declines in democracy following negative shocks, whereas more open regimes exhibit comparatively muted responses. These findings suggest that the average positive relationship between oil shocks and democratization masks important heterogeneity across political regimes and asymmetric responses over the oil price cycle.

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Introduction

Brückner et al. (2012) provide a seminal empirical defense of the “Modernization Hypothesis” within the context of the resource curse. By using international oil price fluctuations as exogenous instruments for transitory income shocks, the authors argue that windfalls in oil-exporting countries lead to significant improvements in democratic institutions. Their theoretical mechanism rests on the idea that higher income increases the opportunity cost for citizens of engaging in political conflict while simultaneously raising demand for greater participation and accountability. In essence, they suggest that when income (instrumented by oil revenue) grows, the resulting prosperity acts as a catalyst for democratic reform.

Our critique centres on the assumptions of linear symmetry and parameter homogeneity embedded in this specification. By using a standard linear panel model, Brückner et al. (2012) implicitly assume that an oil shock affects a consolidated democracy in the same way as it affects a hard autocracy. They also assume that oil price “booms” and “busts” are mirror images of one another, so that a +10% price move carries exactly the same institutional weight as a −10% move.

However, recent scholarship on authoritarian resilience suggests that this restriction is untenable. If the “Autocratic Upgrading” hypothesis holds, dictators do not respond to windfalls by democratizing; they respond by perfecting the machinery of repression and patronage. The literature on Asymmetric Survival Strategies suggests that while windfalls may lead to a temporary “peace” (the co-optation effect), it is the collapse of those windfalls, the busts, that triggers institutional instability and the dilemma between concession and crackdown.

This paper contributes to this debate through two econometric extensions:

A. Endogenous Thresholds. Rather than splitting the sample arbitrarily, we use the threshold model of Hansen (2000), which lets the data identify the institutional boundary at which the effect of oil shocks changes. This speaks to the “Conditional Curse” literature (Torvik, 2009) by testing whether there is a structural break, or tipping point, at which regimes move from being unaffected by oil price shocks to exhibiting the significant positive effect reported by Brückner et al. (2012).

B. Local Projections and Asymmetry. Local projections (Jordà, 2005) allow us to estimate non-linear, dynamic impulse response functions (IRFs). This choice is motivated by the macroeconomic literature on asymmetry, which suggests that institutional responses to negative shocks are often more pronounced and persistent than responses to positive ones. By decomposing shocks into booms and busts, we can test whether the results of Brückner et al. (2012) are a statistical artefact of regime crackdowns during price collapses rather than evidence of modernization during price windfalls.

Literature Review

The relationship between resource endowments, economic development and political liberalization remains one of the most contested frontiers in political science. The literature on this question has divided into two dominant schools of thought: Modernization Theory and the Resource Curse.

The first, pioneered by Lipset (1959) and adopted by Brückner et al. (2012), posits that income is a fundamental driver of democratic transition. Under this view, economic windfalls from diverse sources such as natural resources, rainfall levels (Brückner, 2011), and market-based economic systems increase the opportunity cost of political conflict and empower a rising middle class that demands institutional accountability. Acemoglu et al. (2008) provided a rigorous empirical framework for this, though they famously argued that the income-democracy link is often driven by historical “critical junctures” rather than a contemporaneous causal effect. Brückner et al. (2012) sought to challenge this skepticism by using oil price shocks as an exogenous instrument, arguing that transitory income shocks “cause” democratic improvements.

The second school, the Institutionalist/Resource Curse perspective, argues that the source of income matters more than its volume. The intellectual origins of this view lie in the influential work of Sachs and Warner (1995) and Sachs and Warner (2001), which demonstrates a robust negative correlation between natural resource intensity and economic growth, a finding that launched a generation of inquiry into why resource wealth so often fails to deliver prosperity. Subsequent scholarship extended this logic from growth to governance. Mahdavy (1970) introduced the concept of the “rentier state,” arguing that resource rents free governments from the need to tax citizens, thereby severing the accountability relationship that historically drove democratic demands. Ross (2001) systematized these insights, providing cross-national evidence that oil dependence impedes democratization through three mechanisms: the rentier effect (low taxes dampen pressure for representation), repression spending, and modernization blockage. Haber and Menaldo (2011) challenged this consensus, arguing that the resource-autocracy link disappears once one accounts for historical trends, though their findings have been contested on both measurement and methodological grounds (Andersen and Ross, 2014).

A more nuanced synthesis has emerged from scholars emphasizing institutional conditionality. Mehlum et al. (2006) and Robinson et al. (2006) formalize the idea that the effect of a resource shock depends critically on pre-existing institutions. In their framework, resources are a “blessing” only when institutions are “grabber-proof” (capable of channeling rents toward productive uses) and a “curse” when institutions are weak and rent-seeking predominates. Torvik (2009) further develops this intuition, demon-

strating how the same resource windfall can produce divergent outcomes depending on institutional context.

Heydemann (2007) applied this conditional logic to the political sphere, coining “autocratic upgrading” to describe how resource-rich authoritarian regimes strategically adapt, often using rents to perfect repression and co-optation rather than liberalization. Andersen and Ross (2014) provide systematic evidence that the oil-autocracy link operates primarily through regime survival mechanisms, while Wright et al. (2015) show that oil wealth prolongs authoritarian survival specifically through increased spending on internal security. This conditional approach, which rejects both the unconditional optimism of modernization theory and the deterministic pessimism of early resource curse accounts, provides the theoretical motivation for our study. We posit that the effect of oil shocks on democracy is not constant but contingent on the institutional environment, and further complicated by asymmetric and dynamic effects that the original paper does not consider.

We proceed by first summarizing the baseline specification of Brückner et al. (2012), whose replication is reported in Appendix C, and then introduce our two extensions.

Baseline Specification

Brückner et al. (2012) estimate a two-stage least squares (2SLS) model in which international oil price shocks instrument for GDP growth, allowing them to estimate the causal effect of income on democracy. We reproduce their results using the replication data provided by the authors through the *Review of Economics and Statistics* Dataverse, and use the same dataset for our extensions. The baseline specification is:

$$\text{First Stage: } \Delta \ln GDP_{ct} = \alpha_c + \lambda_t + \pi OilShock_{ct} + \mathbf{X}'_{ct}\rho + v_{ct} \quad (1)$$

$$\text{Second Stage: } \Delta Democracy_{ct} = \alpha_c + \lambda_t + \theta \Delta \widehat{\ln GDP}_{ct} + \mathbf{X}'_{ct}\beta + u_{ct} \quad (2)$$

$$\text{where: } \theta = \frac{\partial \text{Democracy}}{\partial GDP}$$

The critical assumption is that θ is constant across countries and regimes, which is restrictive: a 1% increase in GDP is assumed to have the same effect on democratization whether the country is developed or developing. This rules out heterogeneous treatment effects, a well-known limitation of linear instrumental variable models.

The model rests on two further assumptions that condition the validity and interpretability of the results. The first is monotonicity: oil shocks must affect the GDP of all countries in the same direction. The second is the exclusion restriction: oil

price shocks must affect democracy only through GDP growth, and not through other channels such as rentier state dynamics, conflict or geopolitics.

We successfully replicate the baseline results. The exceptions are Table 13, which we could not replicate because of omissions in the replication package, and a minor discrepancy in Table 14; both are documented in Appendix C.

Extensions

To implement our extensions, we consider two panels: an unbalanced panel of 8,978 observations and a balanced panel of 3,910 observations, which is closer to the set of countries and years used by Brückner et al. (2012). Our extensions yield similar results on both panels. This matters because, as the balance table below shows, the balanced panel used in the original paper differs significantly from the unbalanced panel on key variables such as Polity scores.

Table 1: Summary Table: Unbalanced vs Balanced Panel

	Unbalanced	Balanced	Difference
	Mean (SD)	Mean (SD)	(Balanced – Unbalanced)
Polity2 score	0.381 (7.484)	1.016 (7.640)	0.635***
Change in Polity2	0.085 (1.769)	0.119 (1.823)	0.034
Oil price shock (index)	0.001 (0.017)	0.001 (0.013)	-0.001**
Lagged Polity2	0.295 (7.492)	0.897 (7.653)	0.601***
Oil boom shock	0.002 (0.017)	0.002 (0.012)	-0.001***
Oil bust shock	0.001 (0.005)	0.001 (0.004)	-0.000**
Countries	187	85	
Observations	8976	3910	

Notes: Standard deviations in parentheses. Difference column reports balanced minus unbalanced means. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ from two-sample t -tests of equal means. The balanced panel retains only countries observed in all years with non-missing values for all regression variables.

Many countries drop out of the balanced sample because data on their share of world oil exports are missing. This creates a sample selection problem: the excluded countries tend to be poorer and more autocratic, which biases the balanced panel

towards democracies. Comparing results across both panels therefore provides a more robust test of our hypothesis than the original design allows.

Endogenous Threshold Regression

Sample Splitting and Parametric Threshold Estimation

The baseline specification is linear, homogeneous across countries, additive, and has no regime switching. Threshold methods allow us to study the income-democracy relationship further, to illustrate its dependence on a state variable, and to let the threshold itself vary nonparametrically. The remaining question is whether there is reason to believe that the effect of GDP growth on democracy is nonlinear. Oil may bolster democracy only when institutions are already partially democratic, state capacity is already robust, income exceeds a certain level, oil dependence is sufficiently high, or the regime is not fully autocratic.

A threshold specification is warranted if the following hold:

- i.* Modernization effects are only significant after some income level.
- ii.* Oil-driven growth affects democracies differently than autocracies.
- iii.* Resource rents have heterogeneous effects on autocracies.

This extension tests whether the dataset used by Brückner et al. (2012) contains an endogenous threshold below which the positive income-democracy link changes significantly, as our hypothesis predicts.

Proposed Panel Threshold Version:

$$\text{First Stage: } \Delta \ln GDP_{ct} = \alpha_c + \lambda_t + \pi OilShock_{ct} + \mathbf{X}'_{ct}\rho + v_{ct} \quad (3)$$

$$\begin{aligned} \text{Second Stage (Threshold): } \Delta Democracy_{ct} = & \alpha_c + \lambda_t + \theta_1 \Delta \ln \widehat{GDP}_{ct} \cdot \mathbf{1}[q_{ct} \leq \gamma] \\ & + \theta_2 \Delta \ln \widehat{GDP}_{ct} \cdot \mathbf{1}[q_{ct} > \gamma] + \mathbf{X}'_{ct}\beta + u_{ct} \end{aligned}$$

$$\text{where: } \frac{\partial Democracy}{\partial GDP} = \begin{cases} \theta_1 & \text{if } q_{ct} \leq \gamma \\ \theta_2 & \text{if } q_{ct} > \gamma \end{cases}$$

A key modelling choice is that we use block-bootstrap critical values for the Hansen test rather than asymptotic ones, because the asymptotic distribution of the LR statistic in Hansen (2000) assumes a balanced panel and homoskedastic errors. Since our panel is unbalanced (131 countries with varying time horizons) and subject to cross-country heteroskedasticity, asymptotic critical values are likely too small, risking over-rejection

of the null. The block bootstrap resamples whole country time series, preserving within-country serial correlation, and reconstructs the finite-sample distribution of the LR statistic.

Nonparametric Threshold Estimation

To assess robustness, we also implement the more recent nonparametric threshold regression of Lee and Wang (2023). This model provides a nonparametric check on the Hansen (2000) specification and shows whether the threshold itself varies with a country's existing level of democracy rather than being a single fixed point. While Hansen (2000) assumes a constant threshold parameter γ , Lee and Wang (2023) allow the threshold to be a smooth function of a state variable, formally $\gamma = \gamma(s_{ct})$. This permits the regime boundary to vary continuously with institutional or economic conditions, rather than being fixed at a single value.

In our application, $q_{ct} = PetroShock_{ct}$ is the threshold variable and $s_{ct} = Polity2_{c,t-1}$ is the state variable. The estimable quantity of interest is $\hat{\gamma}(s)$: the level of oil shock at which the democracy response changes regime, as a function of the country's existing democracy level. If $\hat{\gamma}(s)$ is approximately constant, this supports Hansen's assumption. If it varies substantially, Hansen's single-threshold model is misspecified.

The Lee–Wang specification requires that the threshold function $\gamma(s)$ be continuous and sufficiently smooth to allow for valid local kernel estimation. Additionally, the bandwidth sequence must satisfy $b_n \rightarrow 0$ and $nb_n \rightarrow \infty$, ensuring local identification. We further require non-degenerate regimes, $P(q_{ct} \leq \gamma(s_{ct})) \in (0, 1)$, instrument validity $E[OilShock_{ct}u_{ct}] = 0$, and sufficiently weak cross-sectional and temporal dependence for asymptotic theory to apply.

The formal Lee–Wang specification is:

$$\Delta Democracy_{ct} = x'_{ct}\beta_0 + x'_{ct}\delta_0 \cdot \mathbf{1}[q_{ct} \leq \gamma_0(s_{ct})] + u_{ct} \quad (4)$$

where $x_{ct} = PetroShock_{ct}$, so β_0 captures the effect above the threshold and $\beta_0 + \delta_0$ the effect below it. Country and year fixed effects are absorbed by two-way within-transformation. The threshold function $\gamma_0(\cdot)$ is estimated by local constant least squares following Lee and Wang (2023), with bandwidth $b_n = c \cdot n^{-1/2}$ where $c = 2.0$ is selected to ensure sufficient local support given the discrete integer-valued support of s . Pointwise 95% confidence intervals are constructed by inverting the LR statistic, as proposed in that framework.

The identification assumption here is stronger than for Hansen, as the threshold function $\gamma_0(s)$ must be twice continuously differentiable in s . Since *Polity2* takes dis-

crete integer values from -10 to $+10$, this smoothness assumption is an approximation. We treat $\hat{\gamma}(s)$ as a smooth approximation evaluated at a grid of *Polity2* values; any step-function appearance reflects the discrete support of s rather than a model violation, as discussed in Lee and Wang (2023).

Threshold Robustness Check

Having established $\hat{\gamma} = -6$ under the parametric Hansen (2000) framework, we conduct a nonparametric robustness check. Hansen’s method treats the threshold as a single fixed scalar, an assumption that cannot be tested within that framework. If the true regime boundary varies with institutional conditions, Hansen’s point estimate of -6 could be an artefact of forcing a heterogeneous threshold into a single number. The nonparametric framework of Lee and Wang (2023) relaxes this assumption and allows the threshold to be a smooth function of the state variable, providing an independent check on whether Hansen’s constant-threshold result is a feature of the data or a consequence of a parametric restriction.

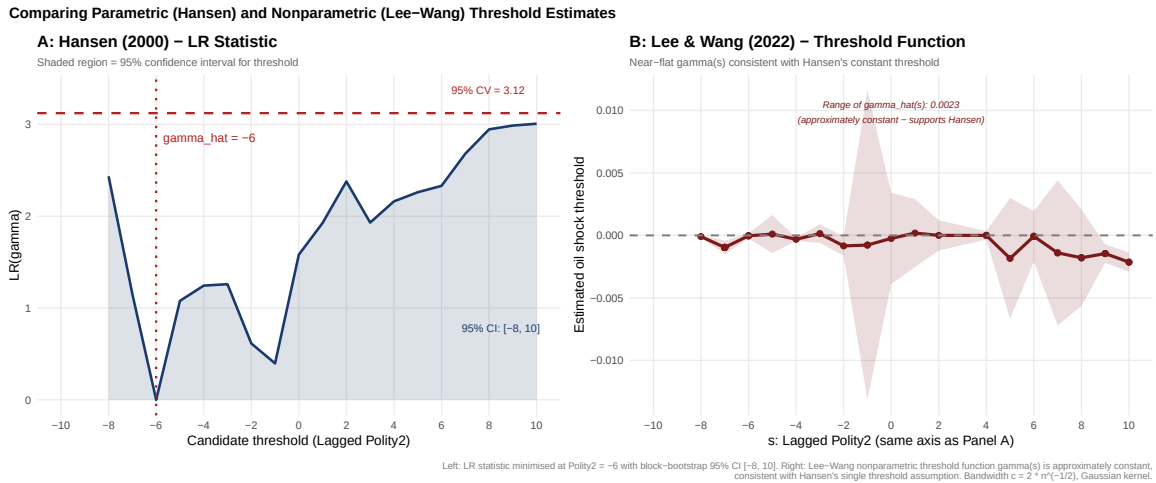


Figure 1: Parametric and Nonparametric Threshold Estimate Comparison. *Left:* Hansen (2000) LR statistic with 95% block-bootstrap confidence interval shaded. The LR curve is minimized at $\hat{\gamma} = -6$. *Right:* Lee and Wang (2023) nonparametric threshold function $\hat{\gamma}(s)$ evaluated at 40 points across the middle 70% of the lagged Polity2 distribution. The near-flat function supports Hansen’s constant-threshold assumption.

The left panel shows the Hansen LR statistic with a 95% block-bootstrap confidence interval; the LR curve reaches its minimum at the estimated threshold of $\hat{\gamma} = -6$. The right panel shows the nonparametric threshold function $\hat{\gamma}(s)$, estimated at 40 evaluation points. The estimated function is nearly constant across the range of s , varying by less than 0.02 oil shock units, which corroborates Hansen’s constant-threshold assumption.

Another key feature is the 95% bootstrap confidence interval for $\hat{\gamma}$, which spans $[-8, 10]$ and indicates a flat residual sum of squares around the minimum. The data therefore support a threshold within this range, but it cannot be pinpointed, a limitation of threshold models discussed in Hansen (2000). This does not undermine the result; it is a limitation of the endogenous threshold framework itself.

The table below reports threshold estimates, regime-specific slope coefficients, and specification details for both methods. The key result is that both methods identify the same qualitative regime structure, with the slope coefficient below the threshold differing in sign from the coefficient above it. The coefficients are not directly comparable in magnitude, since the Hansen estimates are reported in demeaned oil-shock units and the Lee–Wang estimates in standardized units. The comparison therefore rests on the sign, significance and direction of $\hat{\delta}$, which coincide across both models.

Table 2: Comparison of Parametric and Nonparametric Threshold Estimates

	Hansen (2000) Parametric Threshold	Lee & Wang (2022) Nonparametric Threshold
<i>Threshold estimate</i>		
Threshold variable (q)	Lagged Polity2	Oil shock growth
State variable (s)	—	Lagged Polity2
$\hat{\gamma}$ (point estimate)	-6	-0.0008
Range of $\hat{\gamma}(s)$	—	0.0023 (approx. constant)
95% CI	$[-8, 10]$	Pointwise CI in Figure
<i>Regime coefficients</i>		
$\hat{\beta}$ below threshold	-0.5902 (0.6631)	-0.0192 (0.0778)
$\hat{\beta}$ above threshold	7.6363* (4.4193)	0.0195 (0.0122)
$\hat{\delta}$ (difference)	8.2265	0.0004
<i>Specification</i>		
Sample	5380 obs, 131 countries	5380 obs
Fixed effects	Two-way (within)	Two-way (within)
Inference	Block-bootstrap LR	LR inversion (Corollary 1)
Bandwidth	—	$b_n = 2 \cdot n^{-1/2}$ (CV-selected)
Kernel	—	Gaussian

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Hansen coefficients are in oil-shock demeaned units; Lee-Wang in standardised units. Hansen 95% CI from block-bootstrap LR inversion (500 replications, country-block resampling). Lee-Wang 95% CI is pointwise, from LR inversion per Corollary 1 of Lee & Wang (2022).

The threshold estimates show a strong, statistically significant positive relationship between oil price shocks and Polity scores above the threshold, and a small, negative

and statistically insignificant coefficient below it.¹ This result supports our hypothesis of a non-constant relationship that depends heavily on initial levels of democratization, which the original paper does not consider. These findings are consistent with the conditional resource curse literature, in which institutional frameworks significantly alter the response to income shocks. While weak autocracies and democracies see an impetus towards democratization, entrenched autocracies respond differently to identical shocks.

This approach has two advantages. First, combining parametric and nonparametric estimators allows us to independently confirm the existence of a threshold at which the sign and significance of the coefficient change. Second, estimating the threshold endogenously, rather than imposing linear interaction terms, avoids any *ex ante* assumption about where the boundary lies on the Polity scale, which runs from -10 to $+10$, with positive values denoting democracies and negative values denoting autocracies.

Local Projections and Impulse Response Functions

To trace the dynamic response of democracy to oil shocks over time, we use local projections (Jordà, 2005). We prefer local projections to alternatives such as vector autoregressions because they avoid parametric lag structures, which would impose symmetric lag structures across all variables. Local projections remain consistent when the true data generating process (DGP) is not a finite-order VAR, and they accommodate state-dependent interactions naturally: each horizon is estimated as a separate regression, which allows regime-specific coefficients to be estimated:

$$\beta = \begin{bmatrix} \beta_{autocracy}^{boom} & \beta_{autocracy}^{bust} \\ \beta_{democracy}^{boom} & \beta_{democracy}^{bust} \end{bmatrix}$$

We set a horizon of $h = 5$ years, beyond which the sample size falls and efficiency is lost in LP estimation.

We further interact each shock with regime type, defined by whether lagged *Polity2* exceeds the endogenously estimated Hansen threshold of $\hat{\gamma} = -6$ (see the section on Endogenous Threshold Regression). We define countries below the threshold as “hard” autocracies.

This asymmetric decomposition allows us to model how shocks affect different regime types. We hypothesize that booms and busts do not affect autocratic and democratic regimes in the same way: institutionally fragile hard autocracies are likely to respond more sharply to shocks, while the institutional stability of democratic regimes

¹Our results are robust to using the balanced panel; in fact, the differences in the coefficients become larger and more significant on the balanced panel. See Table 4 in the appendix.

allows them to absorb the same shocks with less instability. This approach avoids treating booms and busts symmetrically, which is unlikely given the asymmetric political economy of resource rents.

Formally, at each horizon $h \in \{0, 1, 2, 3, 4, 5\}$, we estimate:

$$\begin{aligned} \Delta_h Democracy_{c,t+h} = & \alpha_c + \lambda_t \\ & + \beta_{aut}^{boom} \cdot Boom_{ct} \cdot \mathbf{1}[q_{ct} \leq \hat{\gamma}] + \beta_{aut}^{bust} \cdot Bust_{ct} \cdot \mathbf{1}[q_{ct} \leq \hat{\gamma}] \\ & + \beta_{dem}^{boom} \cdot Boom_{ct} \cdot \mathbf{1}[q_{ct} > \hat{\gamma}] + \beta_{dem}^{bust} \cdot Bust_{ct} \cdot \mathbf{1}[q_{ct} > \hat{\gamma}] + \mathbf{X}'_{ct}\rho + u_{ct} \end{aligned} \quad (5)$$

where $\Delta_h Democracy_{c,t+h} \equiv Polity2_{c,t+h} - Polity2_{c,t-1}$ is the cumulative change from pre-shock level to horizon h , $Boom_{ct} = \max(0, PetroShock_{ct})$, $Bust_{ct} = |\min(0, PetroShock_{ct})|$, α_c are country fixed effects, λ_t are year fixed effects, and $\mathbf{X}'_{ct}$ includes one lag of the Polity2 change, one lag of each shock component, and the regime dummy. Standard errors are clustered at the country level. The coefficient $\hat{\beta}_{aut}^{boom}$ at horizon h is the impulse response function, interpreted as the expected cumulative change in Polity2 years after a unit boom shock in a hard autocracy, ceteris paribus.

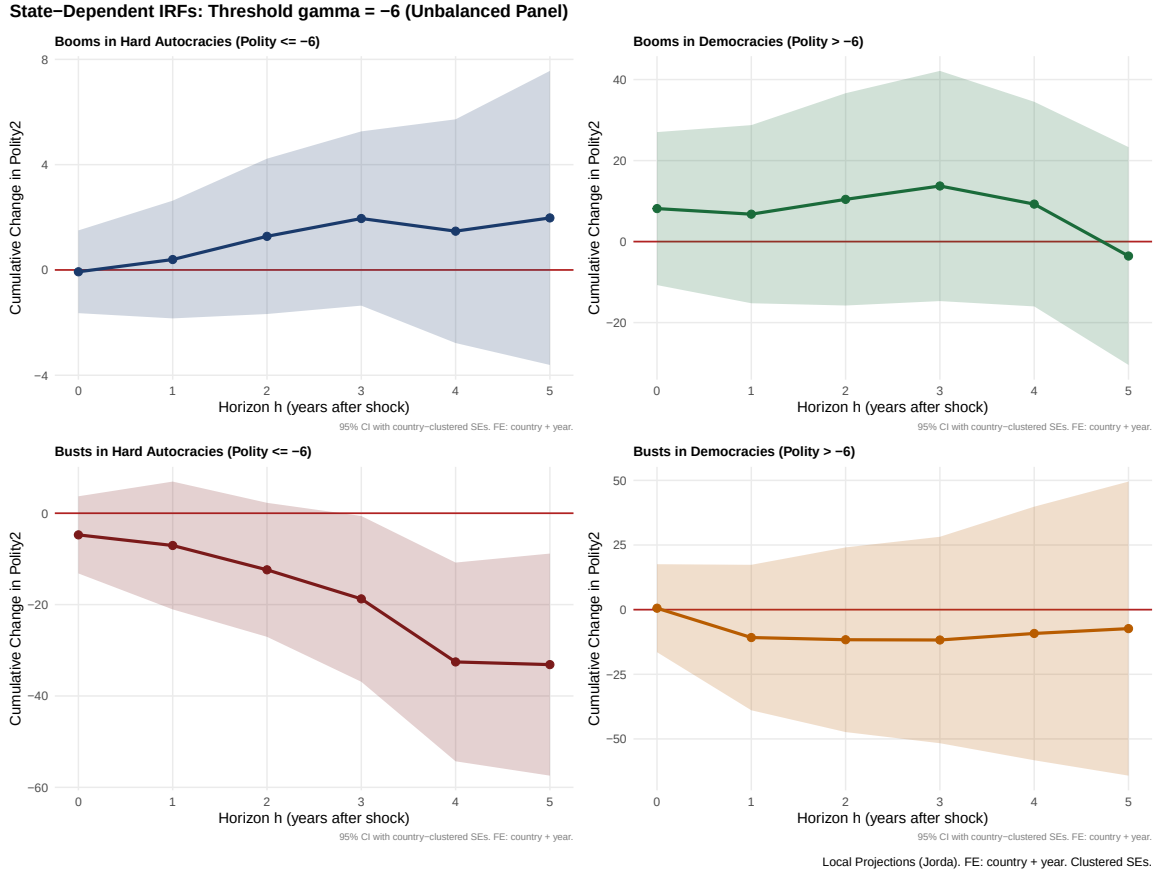


Figure 2: State-Dependent Impulse Response Functions at $\hat{\gamma} = -6$, Unbalanced Panel

The IRFs reveal an asymmetry across shock type and regime, as predicted. In autocracies ($Polity2 \leq -6$), oil booms are associated with a negative change in $Polity2$ initially, followed by small positive changes over the five-year horizon. Oil busts in the same regime show a large decrease in democracy scores. In more open regimes, the responses are less pronounced. Note that the confidence intervals widen significantly at horizons ≥ 3 , reflecting the sample attrition inherent in longer-horizon Local Projection estimates.

These results can be compared with local projections estimated on the balanced panel used in the original paper. The impulse response functions are more statistically significant in this specification, possibly because the exclusion of countries with incomplete time series reduces sample heterogeneity and ensures a constant set of observations across horizons. The stronger statistical significance may reflect this reduced heterogeneity and the greater comparability of the selected countries across horizons, rather than improved identification or reduced bias.

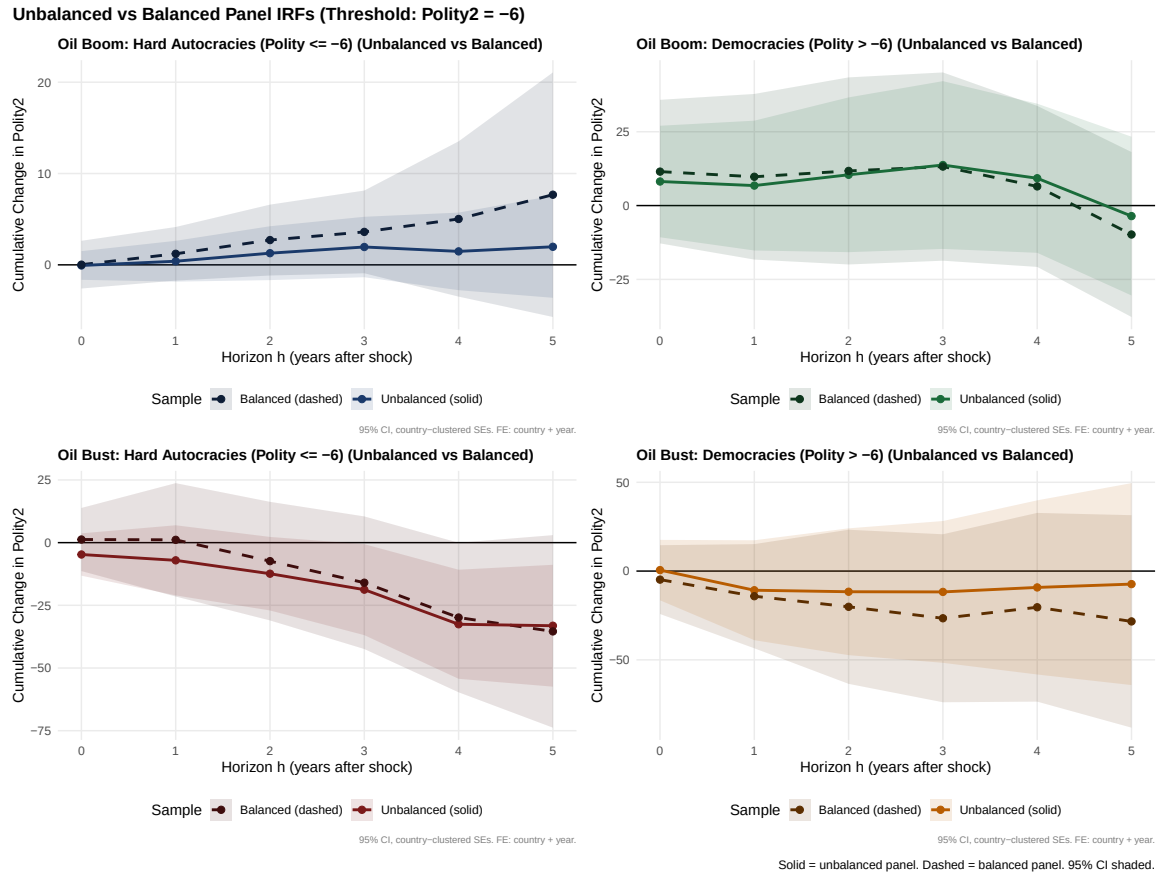
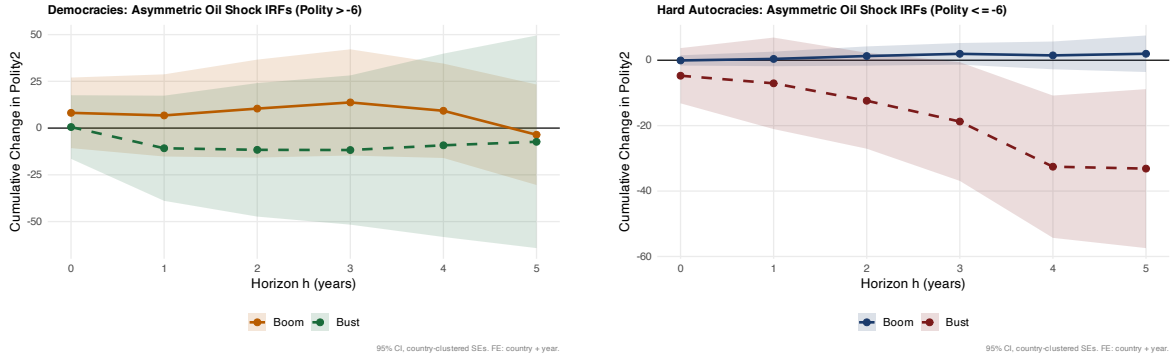


Figure 3: State-Dependent Impulse Response Functions at $\hat{\gamma} = -6$, Balanced vs. Unbalanced Panel

The difference in regime type is visualized as follows:



(a) Oil Shocks in Democracies ($Polity2 > -6$) (b) Oil Shocks in Autocracies ($Polity2 \leq -6$)

Figure 4: Asymmetric Oil Shock IRFs: Democracies vs. Hard Autocracies at $\hat{\gamma} = -6$.

These results are robust to an alternative threshold of $\gamma = 0$, which splits countries at the conventional *Polity2* boundary between autocracy and democracy (see Appendix B).

Comparing the results of the asymmetric local projections delivers further insights. In democracies, Figure 4 shows that positive oil price shocks have a positive contemporaneous effect on democracy scores (consistent with the positive coefficients for shocks in the endogenous threshold extension), which decays over time, while negative shocks have no contemporaneous impact and lead to small reductions that also decay over a five-year horizon. Most importantly, both shock IRFs decay to the same point at the end of the horizon, and their confidence intervals overlap and include zero for the entire period. This is consistent with our hypothesis that stronger democratic institutions are less vulnerable to fluctuations in resource prices.

By contrast, hard autocracies exhibit more pronounced, asymmetric responses, which we attribute to their institutional fragility. The effect of a positive shock is minuscule over the entire period, with the mean of the IRF hugging the zero line, but the response to a bust is a major decline in democracy scores, which can be interpreted as a crackdown on dissent. The picture that emerges is one in which hard autocracies use positive oil shocks to maintain “business as usual,” but rely on repression to counteract the effects of a reduction in income. Our results suggest that, rather than lower incomes reducing the opportunity cost of campaigning for more democratic representation, hard autocracies are able to pre-empt pressure for reform by cracking down on dissent.

Conclusion

This paper replicates and critically extends the findings of Brückner et al. (2012), who posit that transitory positive oil shocks induce democratic improvements, providing empirical backing for the Modernization Hypothesis (Lipset, 1959). Although our baseline replication reproduces the original linear and symmetric specifications, our extensions show that their optimistic conclusion conceals substantial heterogeneity.

By relaxing the assumption of parameter homogeneity, we find that the effects of oil shocks are better described by a “Conditional Curse.” Using an endogenous threshold model (Hansen, 2000; Lee and Wang, 2023), we identify a structural break at a Polity2 score of -6 . This is consistent with the theoretical frameworks of Mehlum et al. (2006) and Robinson et al. (2006): the effect of a natural resource shock is not uniform but depends on the pre-existing institutional landscape. Oil behaves very differently in a consolidated autocracy than in a transitioning or democratic state. The average positive effect reported by Brückner et al. (2012) reflects an implicit average of a strongly positive and significant relationship in democracies and a small, negative and insignificant coefficient in autocracies.

Furthermore, by moving beyond static, symmetric linear specifications to dynamic Local Projections (Jordà, 2005), we uncover pronounced asymmetries in the data. We find no evidence that resource windfalls systematically democratize hard autocracies; rather, they appear to fund the authoritarian status quo through rent distribution and co-optation (Mahdavy, 1970). The positive linear relationship identified by Brückner et al. (2012) is, at least in part, driven by the other side of the economic cycle: resource busts. When oil prices collapse, dictators face an existential threat as their patronage networks dry up. To survive, they substitute co-optation with severe repression, leading to a sharp decline in democratic institutions. Because a symmetric linear specification averages these crackdowns during busts together with the responses to booms, it can produce the appearance of modernization during windfalls.

Ultimately, our findings support the Institutional perspective of the resource curse (Ross, 2001). In consolidated democracies, the dynamic estimates point to an “Immunity Hypothesis,” in which both booms and busts are absorbed as mean-reverting macroeconomic noise. In hard autocracies, by contrast, oil wealth is not a catalyst for democratic awakening but an instrument of regime survival: it sustains the regime when prices are high and is accompanied by repression when prices collapse.

Future research could take these asymmetric effects as given and explore heterogeneity within democratic and autocratic regimes, examining how different institutions produce sharper or more muted responses to resource shocks. A second avenue is to test whether the endogenous threshold we identify also appears when income shocks

other than oil are used, which would strengthen the external validity of our findings. Finally, applying the threshold approach to alternative indicators of democracy would show whether the patterns we document for hard autocracies are robust to the choice of measure.

Appendix

A Endogenous Threshold Regression

Parametric Threshold Estimation

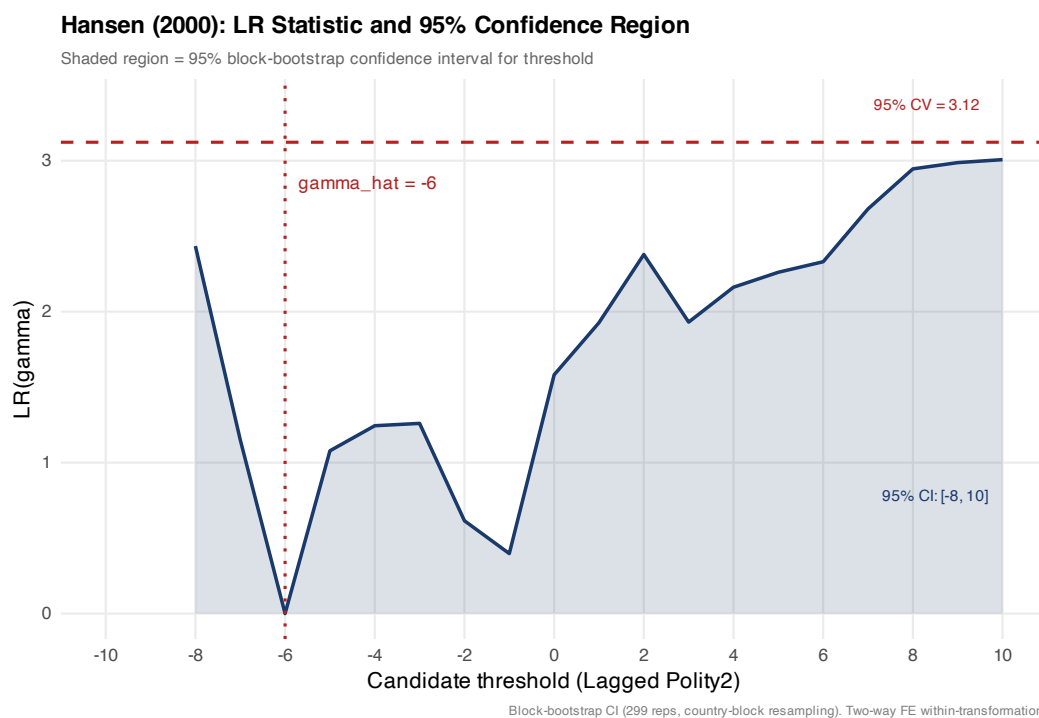
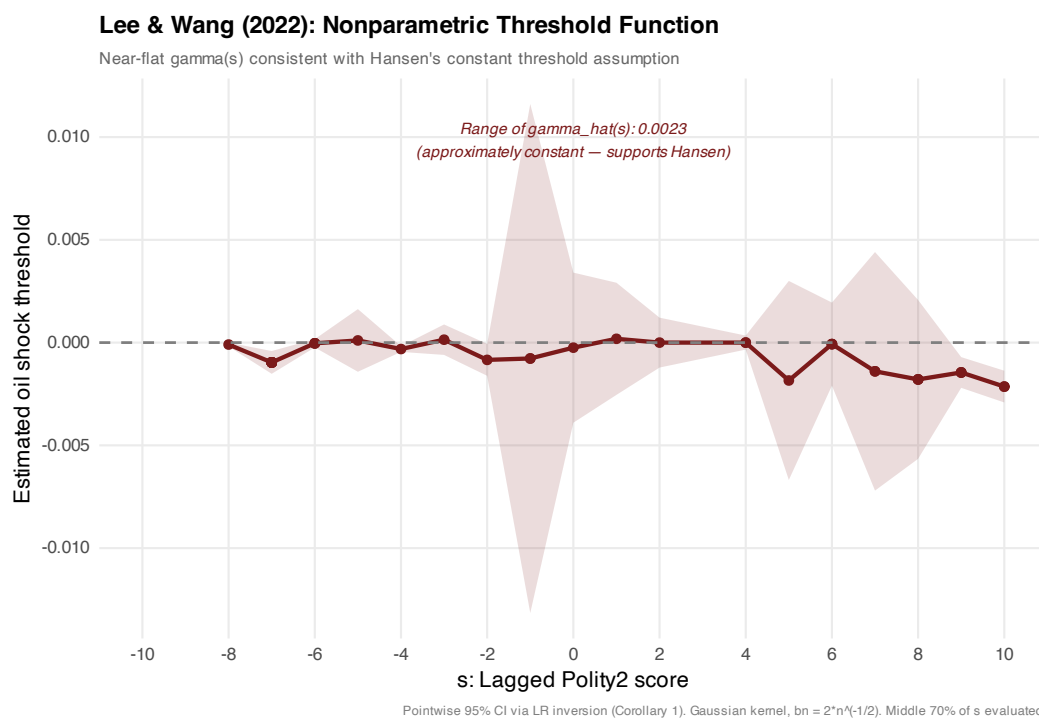


Figure 5: Hansen (2000): LR Statistic and 95% Block-Bootstrap Confidence Region.

Nonparametric Threshold Estimation

Figure 6: Lee and Wang (2023): Nonparametric Threshold Function $\hat{\gamma}(s)$.

Threshold Estimates: Unbalanced vs Balanced Panel

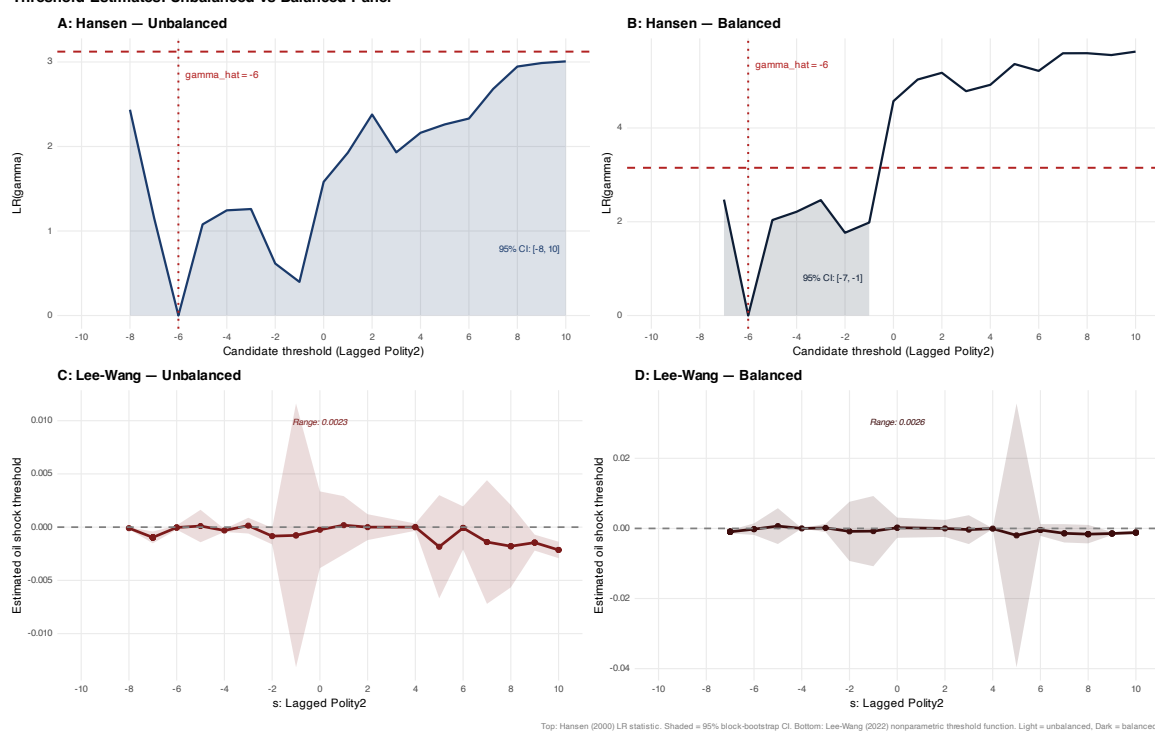


Figure 7: Threshold Estimate Comparison for Balanced and Unbalanced Panels

Table 3: Summary Table: Unbalanced vs Balanced Panel

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Oil price shock (index)	0.001 (0.017)	0.001 (0.013)	-0.001**
Lagged Polity2	0.295 (7.492)	0.897 (7.653)	0.601***
Oil boom shock	0.002 (0.017)	0.002 (0.012)	-0.001***
Oil bust shock	0.001 (0.005)	0.001 (0.004)	-0.000**
Countries	187	85	
Observations	8976	3910	

Notes: Standard deviations in parentheses. Difference column reports balanced minus unbalanced means. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$ from two-sample t -tests of equal means. The balanced panel retains only countries observed in all years with non-missing values for all regression variables.

Table 4: Parametric and Nonparametric Threshold Estimates: Unbalanced and Balanced Panels

	Hansen (2000)		Lee & Wang (2022)	
	Unbalanced	Balanced	Unbalanced	Balanced
<i>Threshold estimate</i>				
$\hat{\gamma}$	-6	-6	Varies	Varies
95% CI	[-8, 10]	[-7, -1]	Fig.	Fig.
<i>Regime coefficients</i>				
$\hat{\beta}$ below threshold	-0.5902 (0.6631)	-1.5826 (1.2579)	-0.0192 (0.0778)	-0.0369 (0.0708)
$\hat{\beta}$ above threshold	7.6363* (4.4193)	13.9238** (6.9924)	0.0195 (0.0122)	0.0279 (0.0221)
$\hat{\delta}$ (difference)	8.2265	15.5064	0.0004	0.0648
<i>Sample</i>				
Observations	5380	3910	5380	3910
Countries	131	85	131	85
Fixed effects	Two-way	Two-way	Two-way	Two-way

Notes: Standard errors in parentheses. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$. Hansen coefficients in demeaned oil-shock units; Lee–Wang in standardised units. Balanced panel retains countries at or above the median number of observations. Bootstrap CI: 299 reps, country-block resampling.

B Local Projection Robustness Checks

Impulse Response Functions (Standard Threshold $\gamma = 0$)

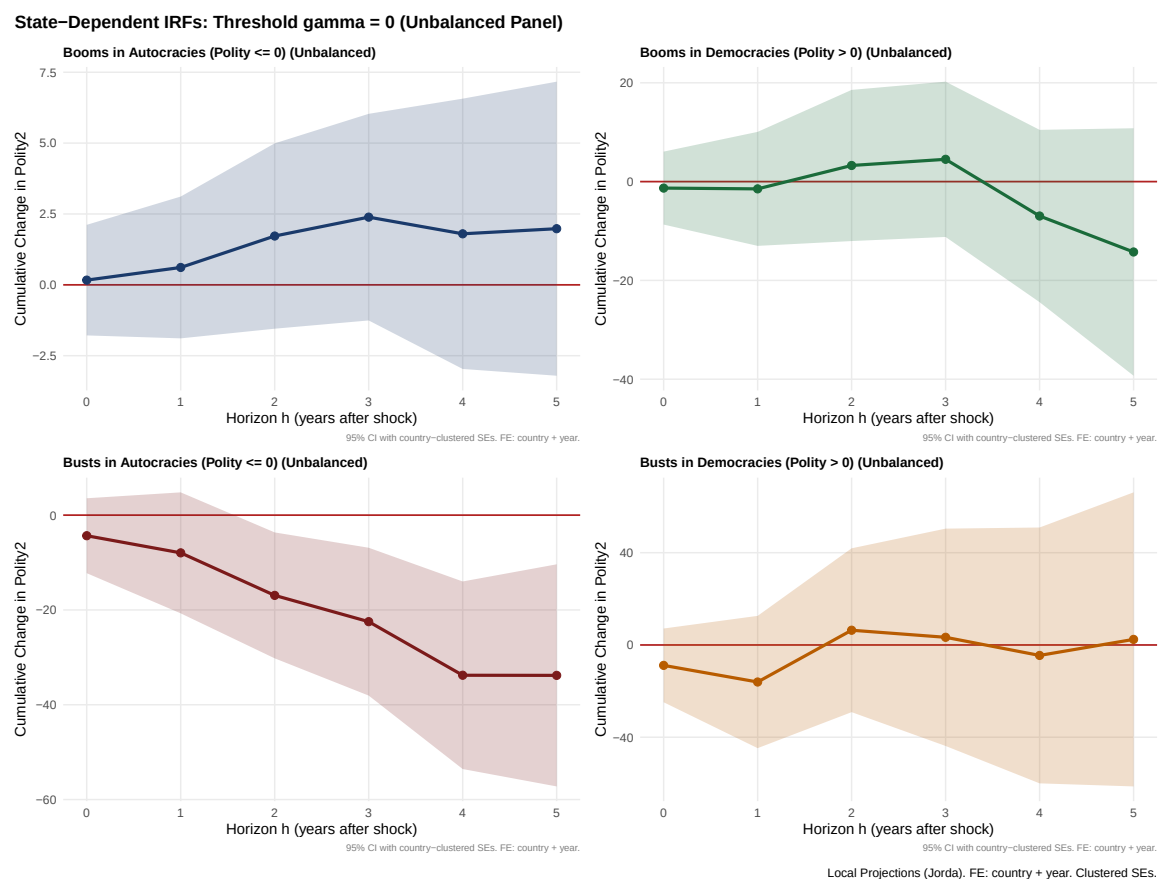
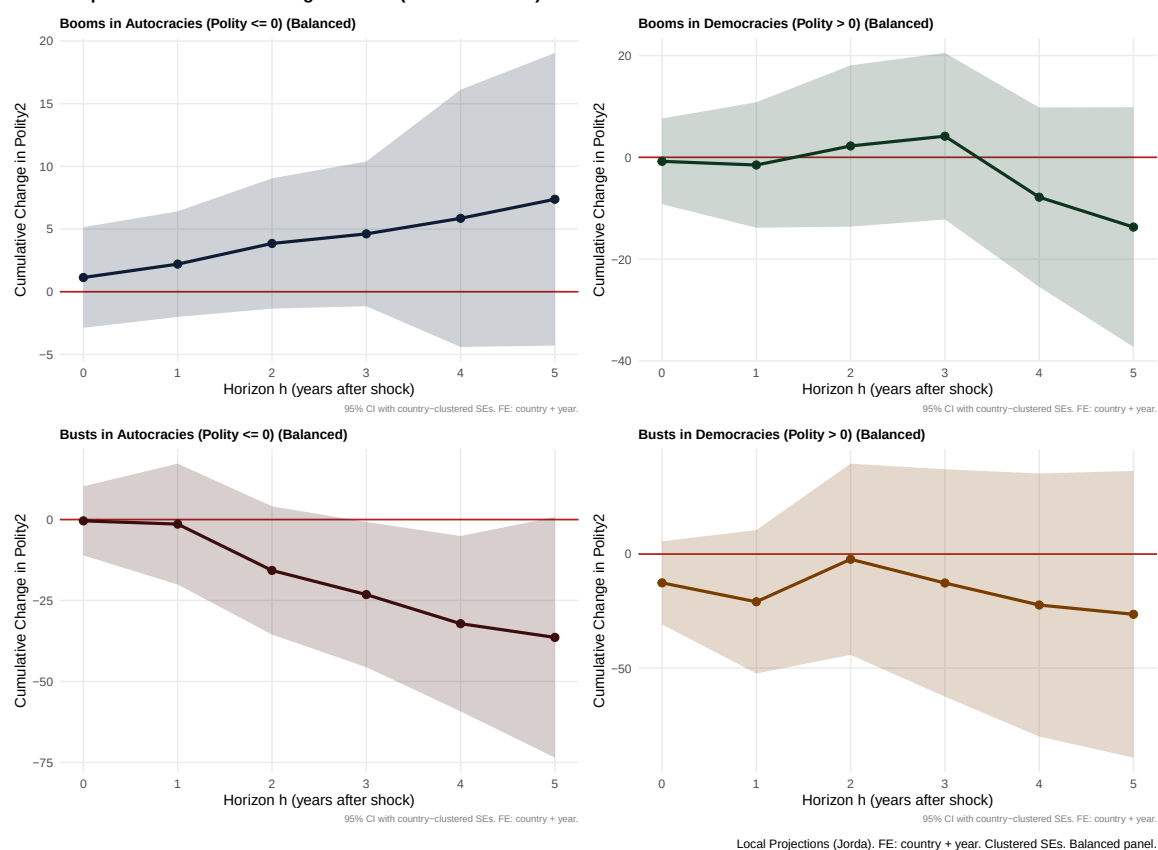
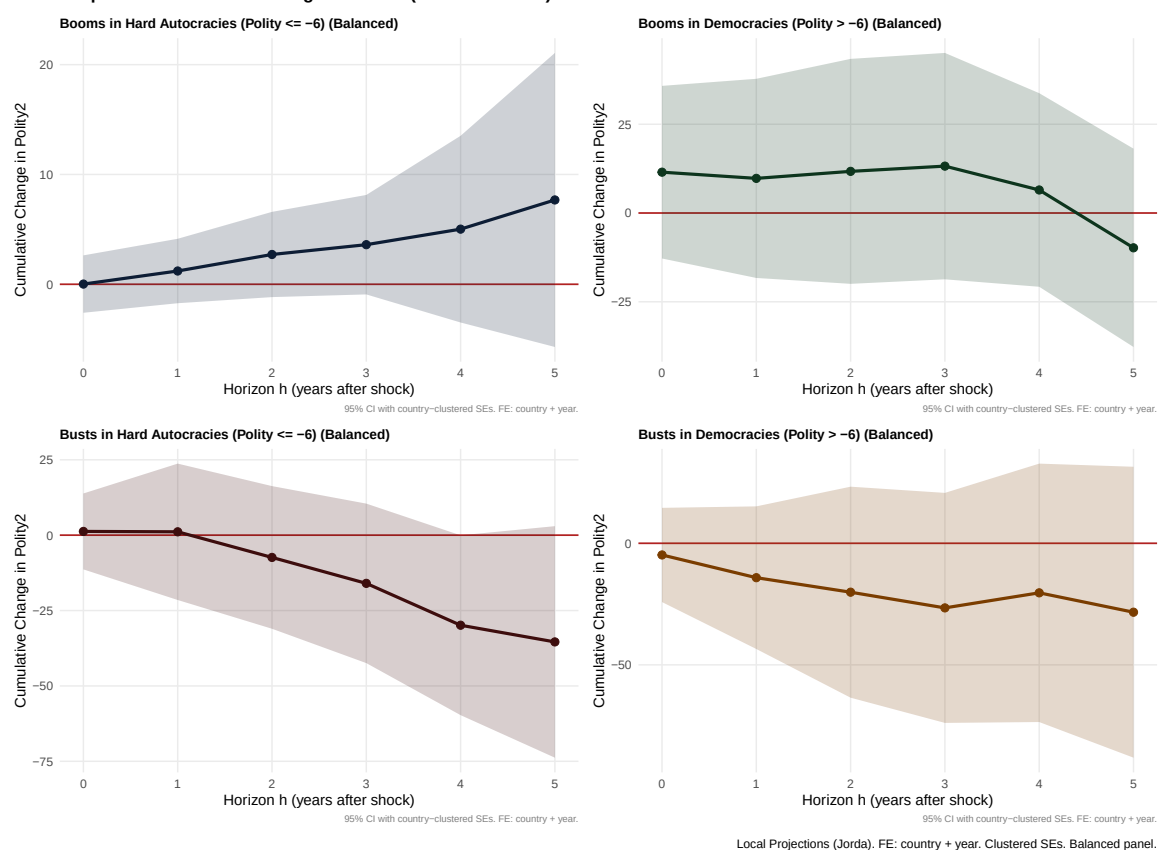


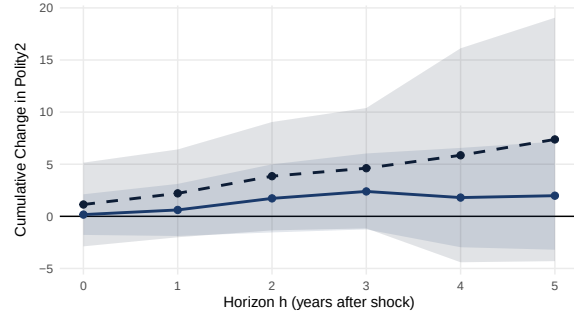
Figure 8: State-Dependent IRFs at $\gamma = 0$ (Unbalanced Panel)

State-Dependent IRFs: Threshold $\gamma = 0$ (Balanced Panel)Figure 9: State-Dependent IRFs at $\gamma = 0$ (Unbalanced Panel)

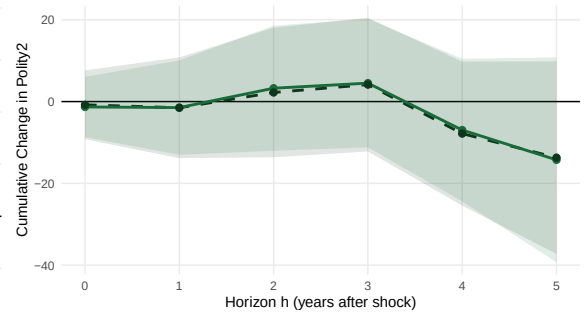
State-Dependent IRFs: Threshold $\gamma = -6$ (Balanced Panel)Figure 10: State-Dependent IRFs at $\gamma = -6$ (Unbalanced Panel)

Unbalanced vs Balanced Panel IRFs (Threshold: Polity2 = 0)

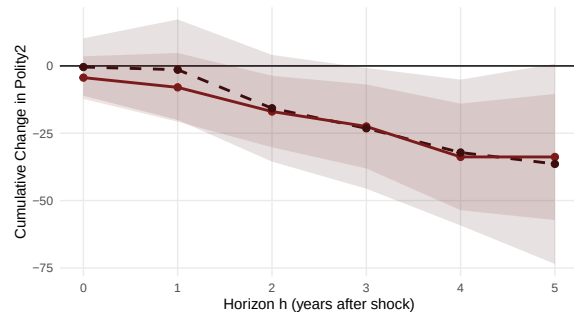
Oil Boom: Autocracies (Polity <= 0) (Unbalanced vs Balanced)



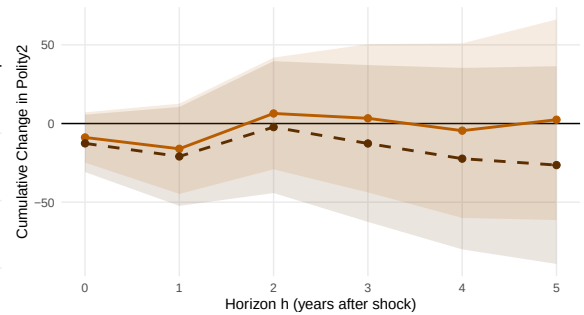
Oil Boom: Democracies (Polity > 0) (Unbalanced vs Balanced)



Oil Bust: Autocracies (Polity <= 0) (Unbalanced vs Balanced)



Oil Bust: Democracies (Polity > 0) (Unbalanced vs Balanced)



Solid = unbalanced panel. Dashed = balanced panel. 95% CI shaded.

Figure 11: State-Dependent IRFs $\gamma = 0$ Balanced vs. Unbalanced Panel

Democracy Oil Shock Comparison ($\gamma = 0$)

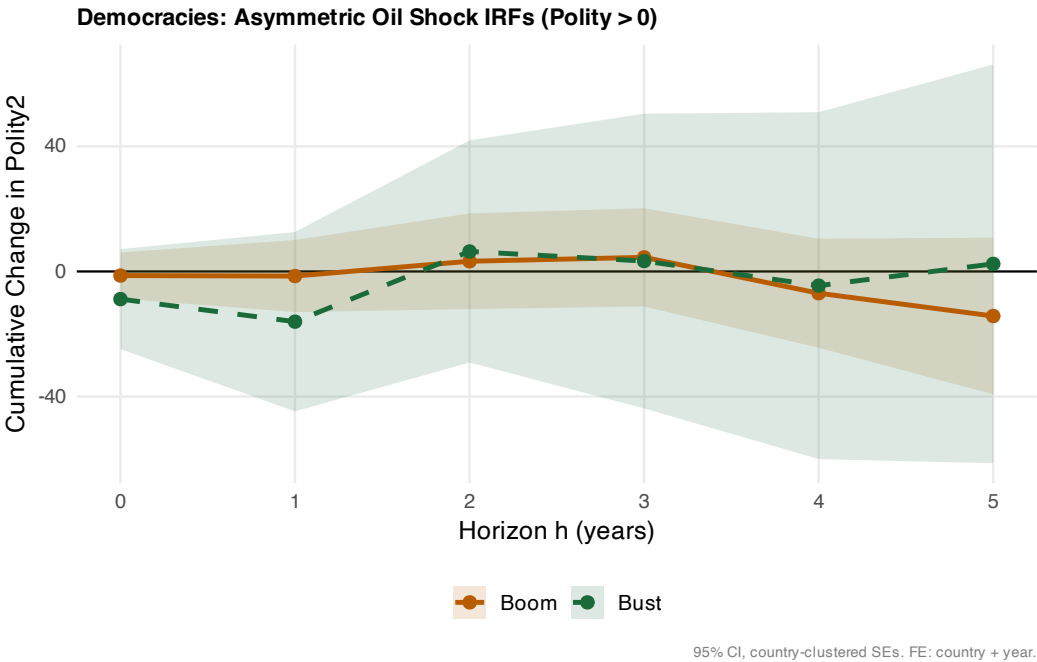


Figure 12: Democracies: Asymmetric Oil Shock IRFs at $\gamma = 0$.

Autocracy Oil Shock Comparison ($\gamma = 0$) Unbalanced Panel

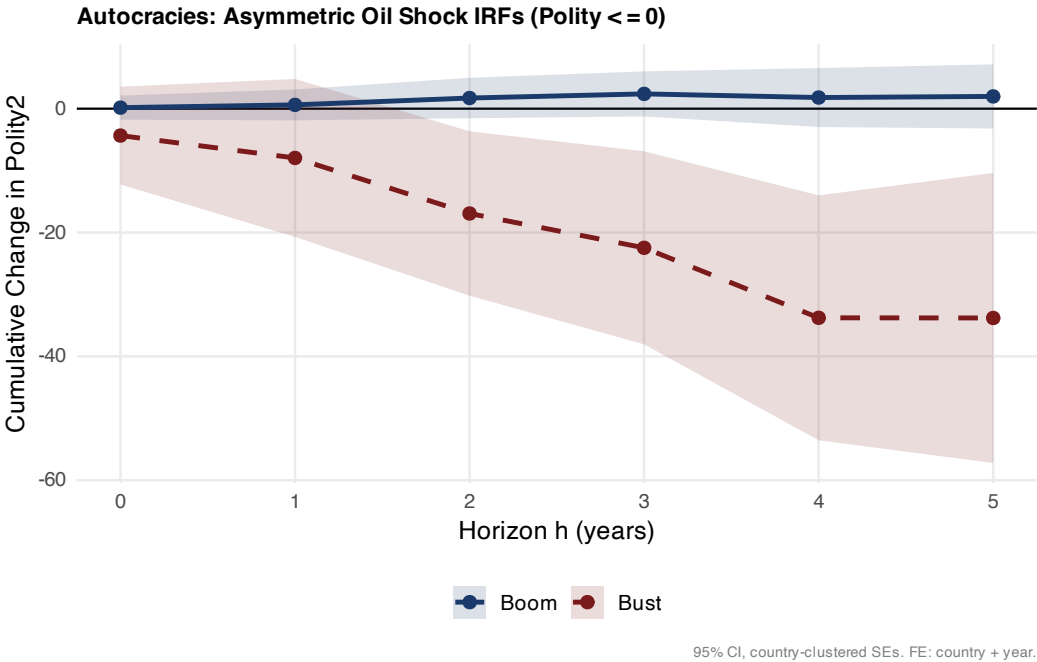


Figure 13: Autocracies: Asymmetric Oil Shock IRFs at $\gamma = 0$.

Table 3 — Oil Price Shocks, Democracy, and Polity Convergence

Panel A: Brückner, Ciccone and Tesei
(2012)

	Δ Polity2		Δ Exconst		Δ Exrec		Δ Polcomp	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	LS	GMM	LS	GMM	LS	GMM	LS	GMM
3-year oil price shock	1.67*** (3.22)	1.70*** (2.93)	0.42** (2.08)	0.36* (1.74)	0.49*** (3.00)	0.48*** (2.87)	0.99*** (3.96)	0.96*** (3.21)
Lagged democracy (Level)	-0.10*** (-8.98)	-0.09*** (-3.12)	-0.12*** (-9.43)	-0.07** (-2.27)	-0.12*** (-9.62)	-0.10*** (-3.40)	-0.10*** (-8.86)	-0.09*** (-3.21)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,782	4,782	4,782	4,782	4,782	4,782	4,782	4,782

Panel B: Replication Output

	Δ Polity2		Δ Exconst		Δ Exrec		Δ Polcomp	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	LS	GMM	LS	GMM	LS	GMM	LS	GMM
3-year oil price shock	1.67*** (3.22)	1.70*** (2.93)	0.42** (2.08)	0.35* (1.74)	0.50*** (3.00)	0.48*** (2.87)	0.99*** (3.96)	0.96*** (3.76)
Lagged democracy (Level)	-0.10*** (-8.98)	-0.10*** (-3.12)	-0.12*** (-9.43)	-0.07** (-2.27)	-0.12*** (-9.62)	-0.10*** (-3.40)	-0.10*** (-8.86)	-0.09*** (-3.21)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,782	4,782	4,782	4,782	4,782	4,782	4,782	4,782

Table 4 — Persistent Effects of Oil Price Shocks on GDP

Panel A: Brückner, Ciccone and Tesei
(2012)

	Δ LnGDP							
	Effect on GDP on Impact ($t \rightarrow t$)		Effect on GDP after Three Years ($t - 3 \rightarrow t$)		Effect on GDP after Five Years ($t - 5 \rightarrow t$)		Effect on GDP after Ten Years ($t - 10 \rightarrow t$)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	LS	GMM	LS	GMM	LS	GMM	LS	GMM
Oil price shock	0.96*** (4.73)	0.94*** (4.90)	0.95*** (5.82)	0.91*** (4.96)	1.00*** (5.32)	0.99*** (5.01)	0.64** (2.25)	0.65** (2.60)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	4,782	4,782	4,625	4,625	4,409	4,409	3,829	3,829

Panel B: Replication Output

	Δ LnGDP							
	Effect on GDP on Impact ($t \rightarrow t$)		Effect on GDP after Three Years ($t - 3 \rightarrow t$)		Effect on GDP after Five Years ($t - 5 \rightarrow t$)		Effect on GDP after Ten Years ($t - 10 \rightarrow t$)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	LS	GMM	LS	GMM	LS	GMM	LS	GMM
Oil price shock	0.96*** (4.73)	0.94*** (4.90)	0.95*** (5.82)	0.91*** (4.96)	1.00*** (5.32)	0.99*** (5.01)	0.64** (2.25)	0.65** (2.60)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	4,782	4,782	4,625	4,625	4,409	4,409	3,829	3,829

Table 5 — Persistent Effects of Oil Price Shocks on Democracy

Panel A: Brückner, Ciccone and Tesei
(2012)

	Δ Polity2							
	Effect on Polity2 on Impact ($t \rightarrow t$)		Effect on Polity2 after Three Years ($t - 3 \rightarrow t$)		Effect on Polity2 after Five Years ($t - 5 \rightarrow t$)		Effect on Polity2 after Ten Years ($t - 10 \rightarrow t$)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	LS	GMM	LS	GMM	LS	GMM	LS	GMM
Oil price shock	1.52*** (3.34)	1.80** (2.37)	4.47*** (2.68)	4.68*** (2.83)	3.34* (1.91)	4.55** (2.04)	4.83** (2.06)	3.77* (1.70)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	4,782	4,782	4,610	4,610	4,390	4,390	3,811	3,811

Panel B: Replication Output

	Δ Polity2							
	Effect on Polity2 on Impact ($t \rightarrow t$)		Effect on Polity2 after Three Years ($t - 3 \rightarrow t$)		Effect on Polity2 after Five Years ($t - 5 \rightarrow t$)		Effect on Polity2 after Ten Years ($t - 10 \rightarrow t$)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	LS	GMM	LS	GMM	LS	GMM	LS	GMM
Oil price shock	1.52*** (3.34)	1.80** (2.37)	4.47*** (2.68)	4.68*** (2.83)	3.34* (1.91)	4.55** (2.04)	4.83** (2.06)	3.77* (1.70)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	4,782	4,782	4,610	4,610	4,390	4,390	3,811	3,811

Table 6 — Oil Price Shocks, Income, and Democracy

Panel A: Brückner, Ciccone and Tesei
(2012)

	Δ Polity2	Δ Exconst	Δ Exrec	Δ Polcomp
	(1)	(2)	(3)	(4)
A: 2SLS Estimates of the Effect of Oil Price Shocks on Democracy through GDP				
Δ LnGDP	4.39*** (3.27)	1.10** (2.09)	1.31*** (3.10)	2.59*** (3.94)
First-stage F -statistic	45	45	45	45
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Number of observations	4,782	4,782	4,782	4,782
B: OLS Estimates of the Effect That GDP Has on Democracy				
Δ LnGDP	0.38 (1.43)	0.21** (2.17)	0.07 (0.67)	0.13 (1.05)
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Number of countries	4,782	4,782	4,782	4,782

Panel B: Replication Output

	Δ Polity2	Δ Exconst	Δ Exrec	Δ Polcomp
	(1)	(2)	(3)	(4)
A: 2SLS Estimates of the Effect of Oil Price Shocks on Democracy through GDP				
Δ LnGDP	4.39*** (3.27)	1.10** (2.09)	1.30*** (3.10)	2.59*** (3.94)
First-stage F -statistic	45	45	45	45
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Number of observations	4,782	4,782	4,782	4,782
B: OLS Estimates of the Effect That GDP Has on Democracy				
Δ LnGDP	0.38 (1.43)	0.20** (2.17)	0.07 (0.67)	0.13 (1.05)
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Number of observations	4,782	4,782	4,782	4,782

Table 7 — Oil Price Shocks, Income, and Democracy: Test of Exclusion Restriction I**Panel A: Brückner, Ciccone and Tesei (2012)**

	$\Delta\text{Polity2}$	$\Delta\text{Exconst}$	ΔExrec	$\Delta\text{Polcomp}$
	(1)	(2)	(3)	(4)
A: IV Is Oil Price Shock and Lagged GDP				
ΔLnGDP	3.97*** (2.63)	1.10** (2.16)	1.22** (2.31)	2.07*** (2.93)
Hansen J , p -value	0.74	0.93	0.85	0.46
First-stage F -statistic	71	71	71	71
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Number of observations	4,759	4,759	4,759	4,759
B: IV Is Lagged GDP				
Three-year oil price shock	0.34 (0.32)	0.03 (0.09)	0.07 (0.19)	0.37 (0.76)
ΔLnGDP	3.58 (1.45)	1.06 (1.27)	1.14 (1.31)	1.65 (1.44)
First-stage F -statistic	40	40	40	40
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Number of countries	4,759	4,759	4,759	4,759

Panel B: Replication Output

	$\Delta\text{Polity2}$	$\Delta\text{Exconst}$	ΔExrec	$\Delta\text{Polcomp}$
	(1)	(2)	(3)	(4)
A: IV Is Oil Price Shock and Lagged GDP				
ΔLnGDP	3.97*** (2.63)	1.10** (2.16)	1.22** (2.31)	2.07*** (2.93)
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
First-stage F -statistic	71	69	71	71
Observations	4,759	4,759	4,759	4,759
B: IV Is Lagged GDP				
Three-year oil price shock	0.34 (0.32)	0.03 (0.09)	0.07 (0.19)	0.37 (0.76)
ΔLnGDP	3.58 (1.45)	1.06 (1.27)	1.14 (1.31)	1.65 (1.44)
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
First-stage F -statistic	40	38	40	41
Observations	4,759	4,759	4,759	4,759

Table 8 — Oil Price Shocks, Income, and Democracy: Test of Exclusion Restriction II**Panel A: Brückner, Ciccone and Tesei (2012)**

	$\Delta\text{Polity2}$	$\Delta\text{Exconst}$	ΔExrec	$\Delta\text{Polcomp}$
	(1)	(2)	(3)	(4)
A: IV Is Oil Price Shock and Lagged Savings Rate				
ΔLnGDP	5.15*** (3.53)	1.34** (2.51)	1.63*** (3.37)	2.69*** (3.86)
Hansen J , p -value	0.30	0.38	0.17	0.91
First-stage F -statistic	25	25	25	25
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	4,731	4,731	4,731	4,731
B: IV Is Lagged Savings Rate				
Oil price shock	-2.26 (-0.81)	-0.62 (-0.75)	-1.09 (-0.99)	-0.13 (-0.11)
ΔLnGDP	10.47 (1.43)	2.79 (1.28)	4.22 (1.48)	3.01 (0.93)
First-stage F -statistic	[0.08]*	[0.11]	[0.08]*	[0.30]
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Number of countries	4,731	4,731	4,731	4,731

Panel B: Replication Output

	$\Delta\text{Polity2}$	$\Delta\text{Exconst}$	ΔExrec	$\Delta\text{Polcomp}$
	(1)	(2)	(3)	(4)
A: IV Is Oil Price Shock and Lagged Savings Rate				
ΔLnGDP	5.15*** (3.53)	1.34** (2.51)	1.63*** (3.37)	2.69*** (3.86)
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
First-stage F -statistic	25	25	25	25
Observations	4,731	4,731	4,731	4,731
B: IV Is Lagged Savings Rate				
ΔLnGDP	10.47 (1.43)	2.79 (1.28)	4.22 (1.48)	3.01 (0.93)
Three-year oil price shock	-2.26 (-0.81)	-0.62 (-0.75)	-1.09 (-0.99)	-0.14 (-0.11)
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
First-stage F -statistic	7	7	7	7
Observations	4,731	4,731	4,731	4,731

Table 9 — Robustness I: Excluding Major Oil Producers and Oil Consumers**Panel A: Brückner, Ciccone and Tesei (2012)**

	$\Delta\text{Polity2}$	$\Delta\text{Exconst}$	ΔExrec	$\Delta\text{Polcomp}$
	(1)	(2)	(3)	(4)
ΔLnGDP	4.05** (2.63)	0.28 (0.36)	1.60*** (2.84)	2.95*** (3.44)
First-stage F -statistic	21	21	21	21
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Number of observations	3,957	3,957	3,957	3,957

Panel B: Replication Output

	$\Delta\text{Polity2}$	$\Delta\text{Exconst}$	ΔExrec	$\Delta\text{Polcomp}$
	(1)	(2)	(3)	(4)
ΔLnGDP	4.05*** (2.63)	0.28 (0.36)	1.60*** (2.84)	2.95*** (3.44)
First-stage F -statistic	22	21	22	22
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Number of observations	3957	3957	3957	3957

Table 10 — Robustness II: Oil Exporters Only

Panel A: Brückner, Ciccone and Tesei (2012)					Panel B: Replication Output				
	$\Delta\text{Polity2}$	$\Delta\text{Exconst}$	ΔExrec	$\Delta\text{Polcomp}$		$\Delta\text{Polity2}$	$\Delta\text{Exconst}$	ΔExrec	$\Delta\text{Polcomp}$
	(1)	(2)	(3)	(4)		(1)	(2)	(3)	(4)
ΔLnGDP	6.91*	2.27*	2.01*	3.65**	ΔLnGDP	6.91*	2.27*	2.02*	3.65**
	(1.95)	(1.77)	(1.75)	(2.38)		(1.95)	(1.77)	(1.75)	(2.38)
First-stage F -statistic	22	22	22	22	First-stage F -statistic	22	22	21	22
Country FE	Yes	Yes	Yes	Yes	Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Year FE	Yes	Yes	Yes	Yes
Number of observations	1,518	1,518	1,518	1,518	Number of observations	1518	1518	1518	1518

Table 11 — Robustness III: Is the Structural Relationship Different for OPEC?

Panel A: Brückner, Ciccone and Tesei (2012)					Panel B: Replication Output				
	$\Delta\text{Polity2}$	$\Delta\text{Exconst}$	ΔExrec	$\Delta\text{Polcomp}$		$\Delta\text{Polity2}$	$\Delta\text{Exconst}$	ΔExrec	$\Delta\text{Polcomp}$
	(1)	(2)	(3)	(4)		(1)	(2)	(3)	(4)
ΔLnGDP	3.30***	0.53	1.12***	2.31***	ΔLnGDP	3.30***	0.53	1.12***	2.31***
	(2.83)	(1.05)	(2.77)	(3.07)		(2.83)	(1.05)	(2.77)	(3.07)
ΔLnGDP^*	1.43	0.74	0.24	0.37	$\Delta\text{LnGDP} \times \text{OPEC}$	1.43	0.74	0.24	0.38
OPEC	(1.11)	(1.32)	(0.70)	(0.60)		(1.11)	(1.32)	(0.70)	(0.60)
First-stage F -statistic	15	15	15	15	First-stage F -statistic	15	15	15	15
Marginal effect in OPEC oil exporters	4.73***	1.27**	1.36***	2.68***	Marginal effect in OPEC oil exporters	4.73***	1.28**	1.36***	2.68***
	(3.20)	(2.16)	(3.03)	(3.97)		(3.20)	(2.16)	(3.03)	(3.97)
Country FE	Yes	Yes	Yes	Yes	Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Year FE	Yes	Yes	Yes	Yes
Number of observations	4,782	4,782	4,782	4,782	Number of observations	4782	4782	4782	4782

Table 12 — Robustness IV: Time-Period Sample Split

Panel A: Brückner, Ciccone and Tesei (2012)					Panel B: Replication Output				
	$\Delta\text{Polity2}$	$\Delta\text{Exconst}$	ΔExrec	$\Delta\text{Polcomp}$		$\Delta\text{Polity2}$	$\Delta\text{Exconst}$	ΔExrec	$\Delta\text{Polcomp}$
	(1)	(2)	(3)	(4)		(1)	(2)	(3)	(4)
A: Pre-1987 Period					A: Pre-1987 Period				
ΔLnGDP	2.60*	0.73	0.64	1.50**	ΔLnGDP	2.60*	0.73	0.64	1.50**
	(1.87)	(1.35)	(1.50)	(2.27)		(1.87)	(1.35)	(1.50)	(2.27)
First-stage F -statistic	61	61	61	61	First-stage F -statistic	61	61	61	61
Country FE	Yes	Yes	Yes	Yes	Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Year FE	Yes	Yes	Yes	Yes
Number of observations	2,419	2,419	2,419	2,419	Number of observations	2419	2419	2419	2419
B: Post-1987 Period					B: Post-1987 Period				
ΔLnGDP	2.57**	0.04	0.98*	2.33**	ΔLnGDP	2.57**	0.04	0.99*	2.34**
	(1.96)	(0.07)	(1.93)	(2.54)		(1.96)	(0.07)	(1.93)	(2.54)
First-stage F -statistic	22	22	22	22	First-stage F -statistic	22	22	22	22
Country FE	Yes	Yes	Yes	Yes	Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Year FE	Yes	Yes	Yes	Yes
Number of observations	2,363	2,363	2,363	2,363	Number of observations	2363	2363	2363	2363

Note on Table 13 (Robustness V). We could not replicate Table 13 because of omissions in the replication package. The table relies on an instrument weighted by each country's net-export GDP share, fixed at 1970 levels. This 1970 share variable is missing from the master `data.dta` file, and the only standalone share variable provided (`petroleum_worldshare`) measures a country's share of global production rather than

its domestic reliance on oil in GDP. An instrument reconstructed from this variable fails the weak-instrument test ($F \approx 19.68$, compared with 59.0 in the original), indicating that it is not a valid proxy. Without the original 1970 data, it is also not possible to reproduce the authors' casewise deletion, which restricts their sample to exactly 3,390 observations. We flag this as a limitation of the replication package.

Table 14 — Robustness VI: Including Interregnum Periods and Using Transition Indicators

Panel A: Brückner, Ciccone and Tesei (2012)					Panel B: Replication Output				
	Democratic Transition	Autocratic Reserval	Δ Polity2	Δ Regtrans	Democratic Transition	Autocratic Reversal	Δ Polity2	Δ Regtrans	
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	
$\Delta \ln \text{GDP}$	0.36*** (2.72)	-0.57 (-1.52)	3.80** (2.62)	0.70* (1.89)	0.36*** (2.72)	-0.57 (-0.90)	3.80*** (2.62)	0.70* (1.89)	
First-stage F -statistic	38	12	48	48	39	14	49	48	
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Number of observations	2,649	2,371	5,020	5,020	2649	2369	5020	5020	

Minor discrepancies in the number of observations and coefficients for Table 14, column (2) relative to the original paper are due to differences in singleton group handling between Stata versions. The original authors used Stata 10, whereas more recent versions (in our case Stata 19.5) automatically drop singleton groups when running `xtivreg2`. As indicated by the warning

Warning - singleton groups detected. 2 observation(s) not used.

this results in two fewer observations in our regression compared to the original paper.

Table 15 — Robustness VII: Alternative Democracy Indicators

Panel A: Brückner, Ciccone and Tesei (2012)					Panel B: Replication Output				
	Δ Political Rights	Δ FH Indicator	Przeworski et al. (2000)	Papaioannou and Siourounis (2008b)	Δ Political Rights	Δ FH Indicator	Przeworski et al. (2000)	Papaioannou and Siourounis (2008b)	
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	
$\Delta \ln \text{GDP}$	1.92*** (3.54)	0.78** (1.98)	0.26*** (2.64)	0.14*** (2.71)	1.92*** (3.54)	0.78** (1.98)	0.26*** (2.64)	0.14*** (2.71)	
First-stage F -statistic	34	34	31	29	34	35	31	29	
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Number of observations	4,263	4,263	4,573	4,660	4263	4263	4573	4660	

Table 16 — Robustness VIII: World Bank GDP Data

Panel A: Brückner, Ciccone and Tesei (2012)

	$\Delta\text{Polity2}$	$\Delta\text{Political Rights}$	$\Delta\text{FH Indicator}$	Przeworski et al. (2000)	Papaioannou and Siourounis (2008b)
	(1)	(2)	(3)	(4)	(5)
ΔLnGDP	9.82 (1.31) [0.07]*	5.58 (1.78)* [0.00]***	2.90 (1.49) [0.01]***	0.89 (1.59) [0.05]**	0.32 (1.29) [0.12]
First-stage F -statistic	8	8	8	9	7
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Number of observations	4,507	3,852	3,852	4,029	4,154

Panel B: Replication Output

	$\Delta\text{Polity2}$	$\Delta\text{Political Rights}$	$\Delta\text{FH Indicator}$	Przeworski et al. (2000)	Papaioannou and Siourounis (2008b)
	(1)	(2)	(3)	(4)	(5)
ΔLnGDP	9.82 (1.31) [0.08*]	5.59 (1.78*) [0.00***]	2.89 (1.49) [0.01**]	0.89 (1.50) [0.05**]	0.32 (1.29) [0.12]
First-stage F -statistic	8	8	8	9	7
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Number of observations	4507	3852	3852	4029	4154

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